



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- **Summary of methodologies**
 - Data Collection through API
 - Data Collection with Web Scraping
 - Data Wrangling
 - Exploratory Data Analysis with SQL
 - Exploratory Data Analysis with Data Visualization
 - Interactive Visual Analytics with Folium
 - Machine Learning Prediction
- **Summary of all results**
 - Exploratory Data Analysis result
 - Interactive analytics in screenshots
 - Predictive Analytics result

Introduction

Project background and context:

- Space travel costs decreasing due to private companies (SpaceX, Blue Origin, Virgin Galactic).
- SpaceX's reusable Falcon 9 reduces costs to \$62M (vs \$165M+ for competitors).
- Goal: Predict first-stage landing success to estimate launch costs.
- Value: Enables competitors to benchmark against SpaceX.

Problems you want to find answers

- Which key factors influence the successful landing of a rocket?
- How different features interact to affect the likelihood of a successful landing?
- What operating conditions are necessary to ensure a reliable and successful landing program?

Section 1

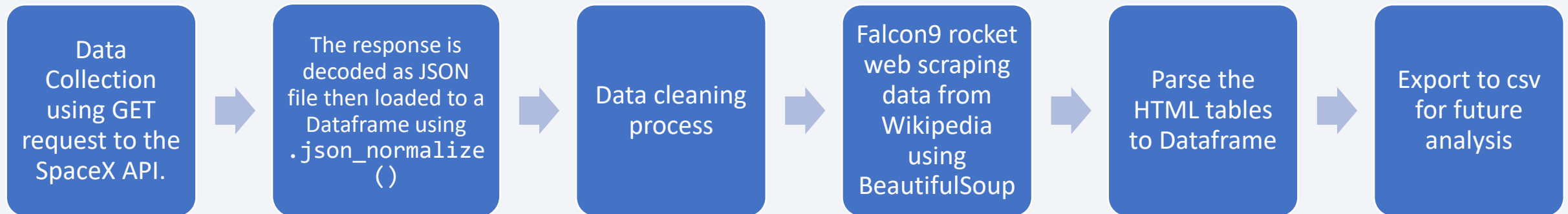
Methodology

Methodology

- Data collection methodology:
 - Data was collected using SpaceX API and web scraping.
- Perform data wrangling
 - One-hot encoding was applied to categorical features
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - How to build, tune, evaluate classification models

Data Collection

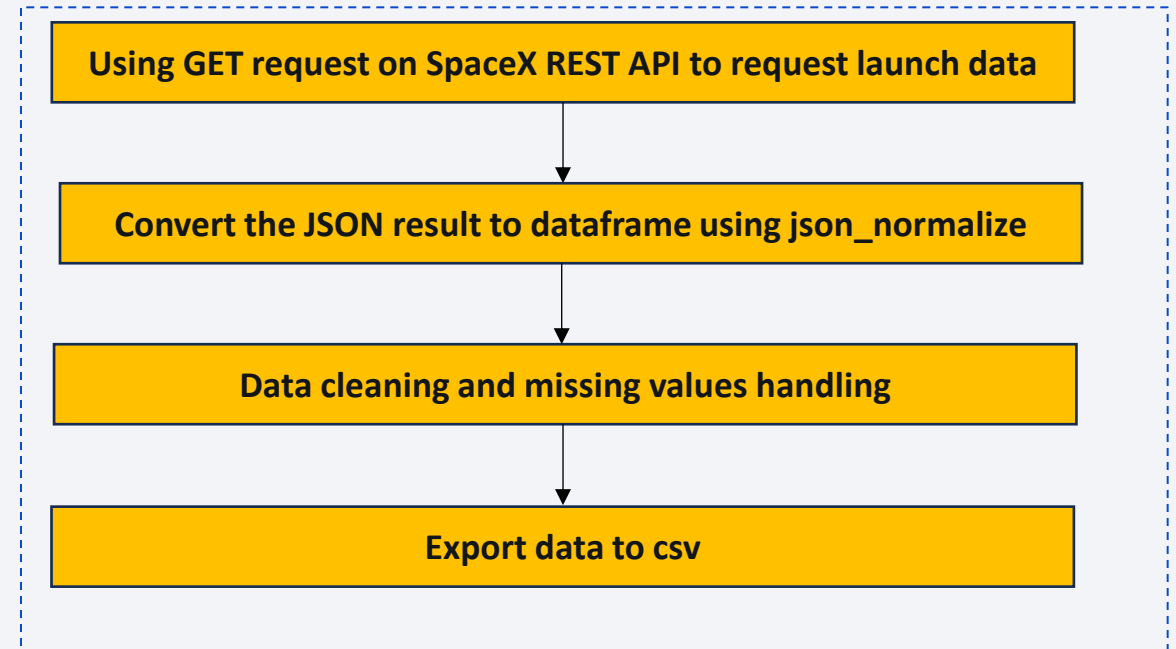
- How were the data sets collected?



Data Collection – SpaceX API

- Using the SpaceX REST API to collect data, then cleaning the requested data with some wrangling and formatting and export it to a csv.
- GitHub URL of the notebook is below:

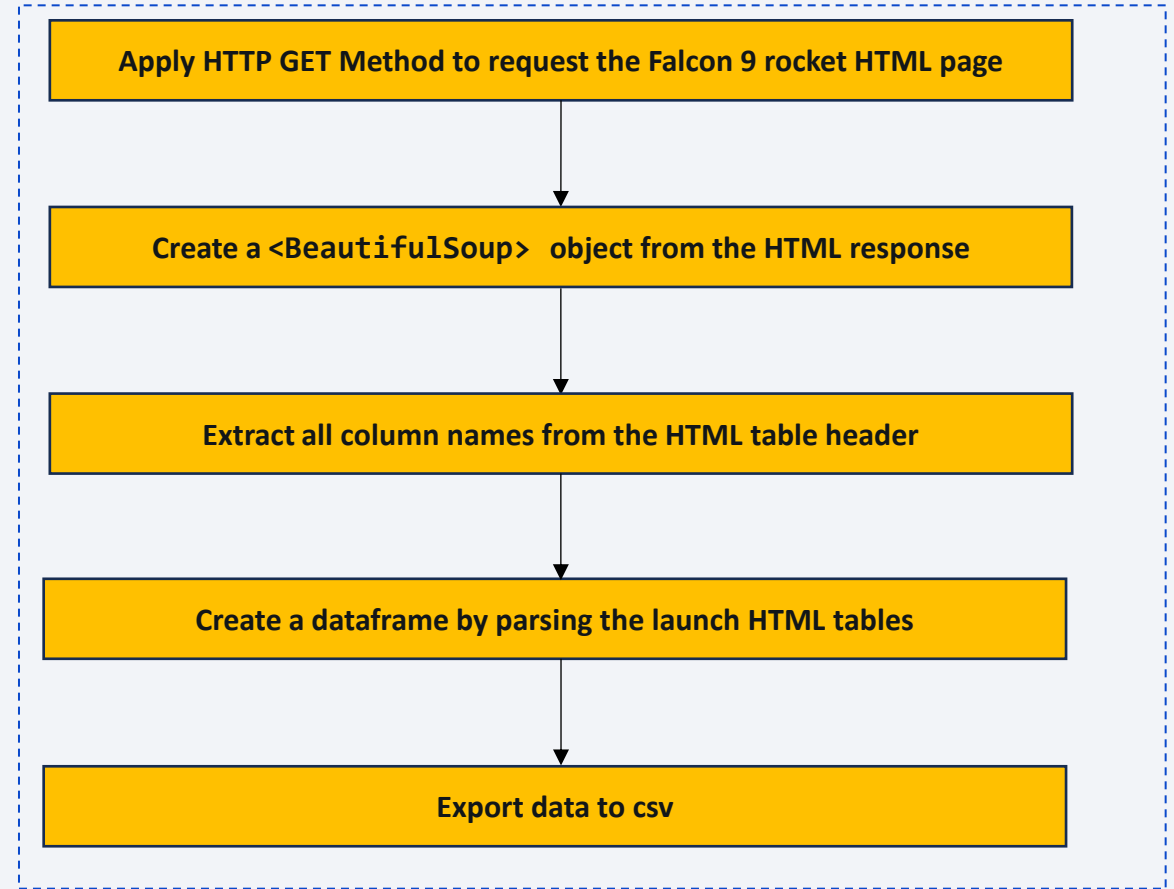
https://github.com/EladdadMahdi/IBM-Applied-Data-Science-Capstone/blob/dev/notebooks/Module_01/1_data-collection-api.ipynb



Data Collection - Scraping

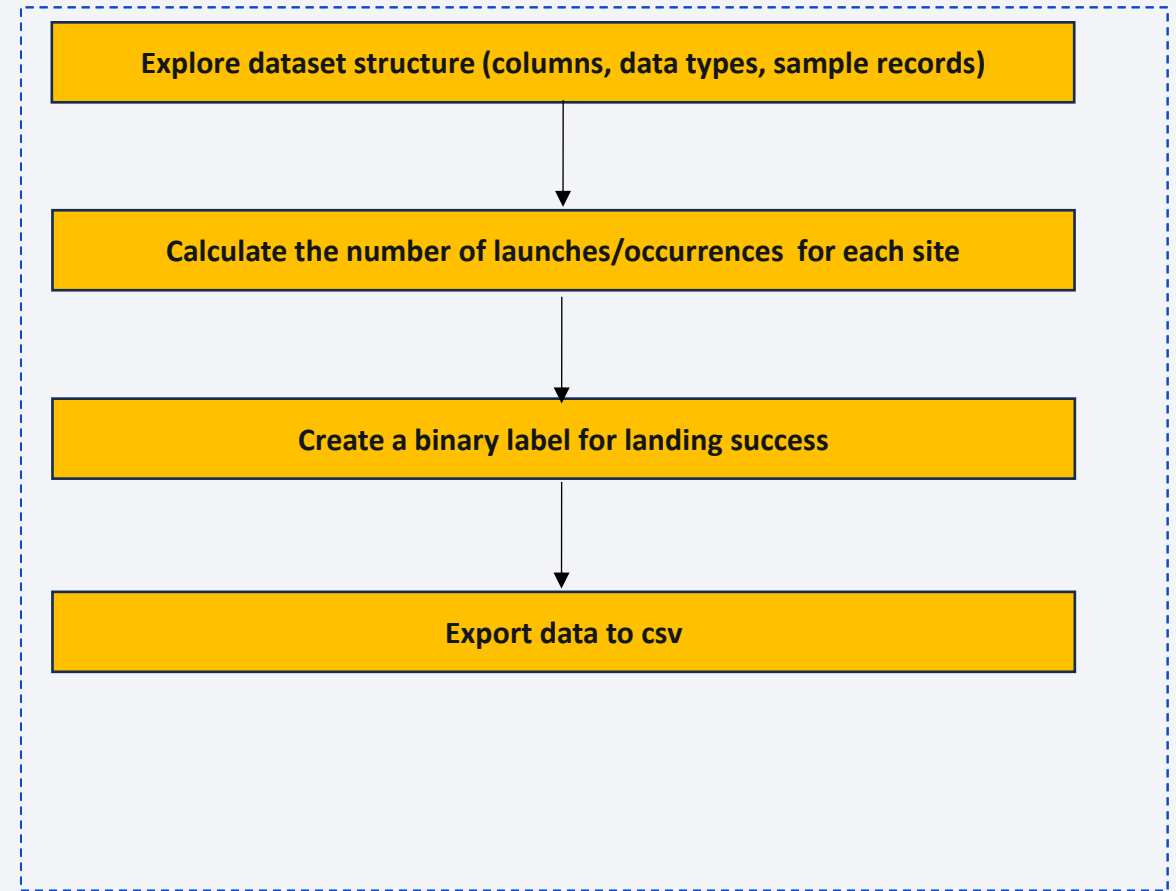
- Web scraping Falcon 9 launch records using BeautifulSoup
- The HTML tables then are parsed to Pandas Dataframe
- GitHub URL of the notebook is below:

https://github.com/EladdadMahdi/BM-Applied-Data-Science-Capstone/blob/dev/notebooks/Module_01/2_data-webscraping.ipynb



Data Wrangling

- Performed exploratory data analysis and determined the training labels.
- Calculated the number of launches at each site, and the number and occurrence of each orbits
- Created landing outcome label from outcome column and exported the results to csv.
- GitHub URL of the notebook is below:
https://github.com/EladdadMahdi/IBM-Applied-Data-Science-Capstone/blob/dev/notebooks/Module_01/3_data-wrangling.ipynb



EDA with Data Visualization

- In this part we explore the data by visualizing the relationship between number and launch Site, payload and launch site, success rate of each orbit type, flight number and orbit type, the launch success yearly trend. Different types of plot are used to represent trends depending on metrics.
- GitHub URL of the notebook is below :

[https://github.com/EladdadMahdi/IBM-Applied-Data-Science-Capstone/blob/dev/notebooks/Module 02/5_eda_dataviz.ipynb](https://github.com/EladdadMahdi/IBM-Applied-Data-Science-Capstone/blob/dev/notebooks/Module%205_eda_dataviz.ipynb)

EDA with SQL

1. In this part of the project the SpaceX dataset is loaded into a DB2 database to extract insights from the dataset we apply EDA with SQL using the queries below:

- The names of unique launch sites in the space mission.
 - The total payload mass carried by boosters launched by NASA (CRS)
 - The average payload mass carried by booster version F9 v1.1
 - The total number of successful and failure mission outcomes
 - The failed landing outcomes in drone ship, their booster version and launch site names.
- GitHub URL of the notebook is below :

[https://github.com/EladdadMahdi/IBM-Applied-Data-Science-Capstone/blob/dev/notebooks/Module 02/4_eda_sql_sqllite.ipynb](https://github.com/EladdadMahdi/IBM-Applied-Data-Science-Capstone/blob/dev/notebooks/Module%202/4_eda_sql_sqllite.ipynb)

Build an Interactive Map with Folium

- I marked all launch sites, and added map objects such as markers, circles, lines to mark the success or failure of launches for each site on the folium map.
- I encoded the feature launch outcomes using binary labels: “0” for failure and “1” for success.
- Using the color-labeled marker clusters, I identified which launch sites have relatively high success rate.
- Finally, I calculated distances between a launch site to its proximities. In order to answer the questions below :
 - Are launch sites in close proximity to railways? highways? coastline?
 - Do launch sites keep certain distance away from cities?
- GitHub URL of the notebook is below :

https://nbviewer.org/github/EladdadMahdi/IBM-Applied-Data-Science-Capstone/blob/dev/notebooks/Module_03/6_Interactive_Visual_Analytics_With_Folium.ipynb

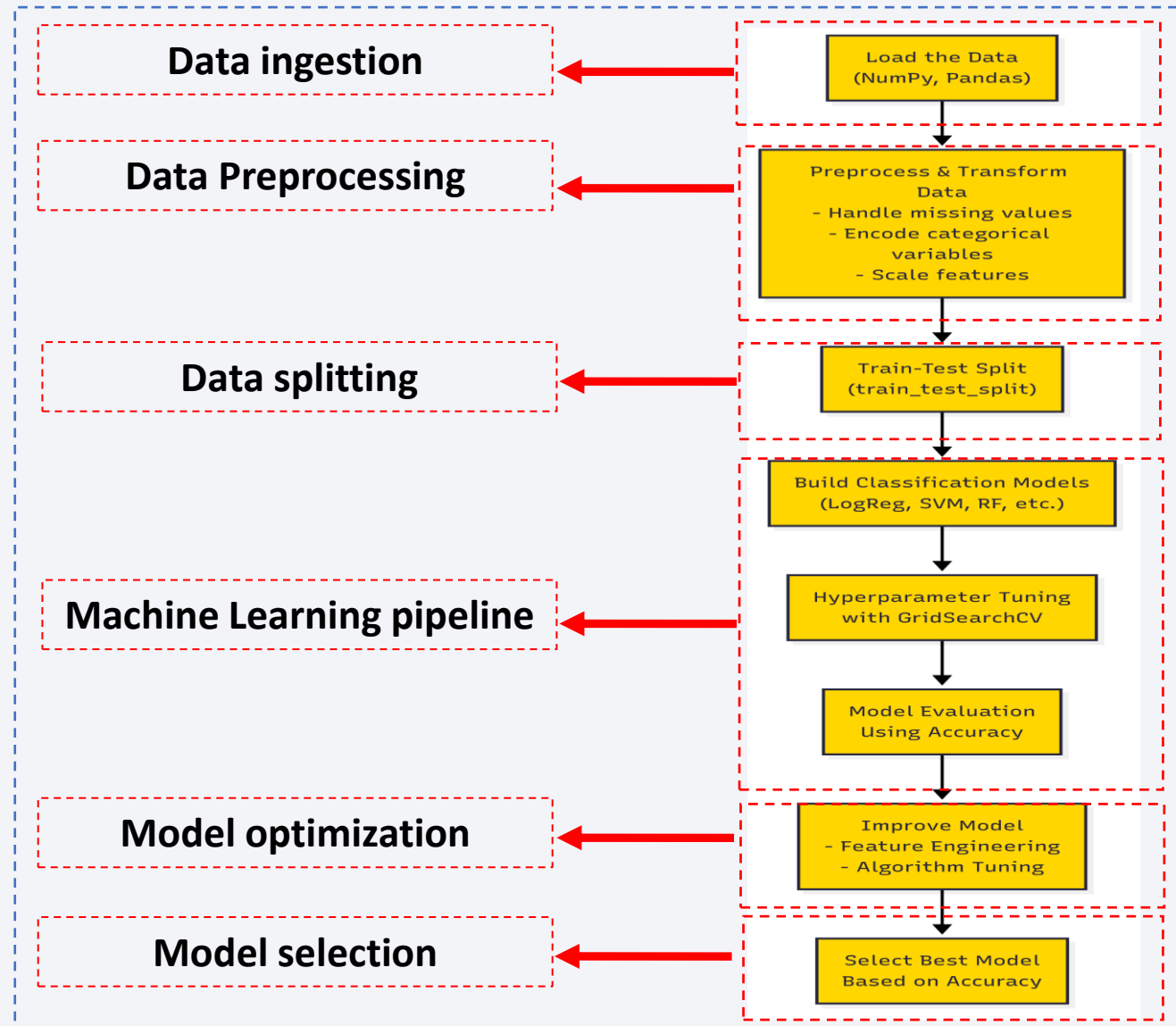
Build a Dashboard with Plotly Dash

- Using the Plotly Dash framework, I built an interactive web dashboard.
- The Pie plot chart showing the total number of launches for each launch sites, pie chart here is the right choice for categorical data each launch site is a category).
- To understanding if payload mass affects launch success. I plotted a scatter graph showing the relationship between the outcome (success:1/Failure:0) and Payload Mass (Kg) for the different booster version.
- GitHub URL of the notebook is below :
- https://github.com/EladdadMahdi/IBM-Applied-Data-Science-Capstone/blob/dev/dashbord/spacex_dash_app.py

Predictive Analysis (Classification)

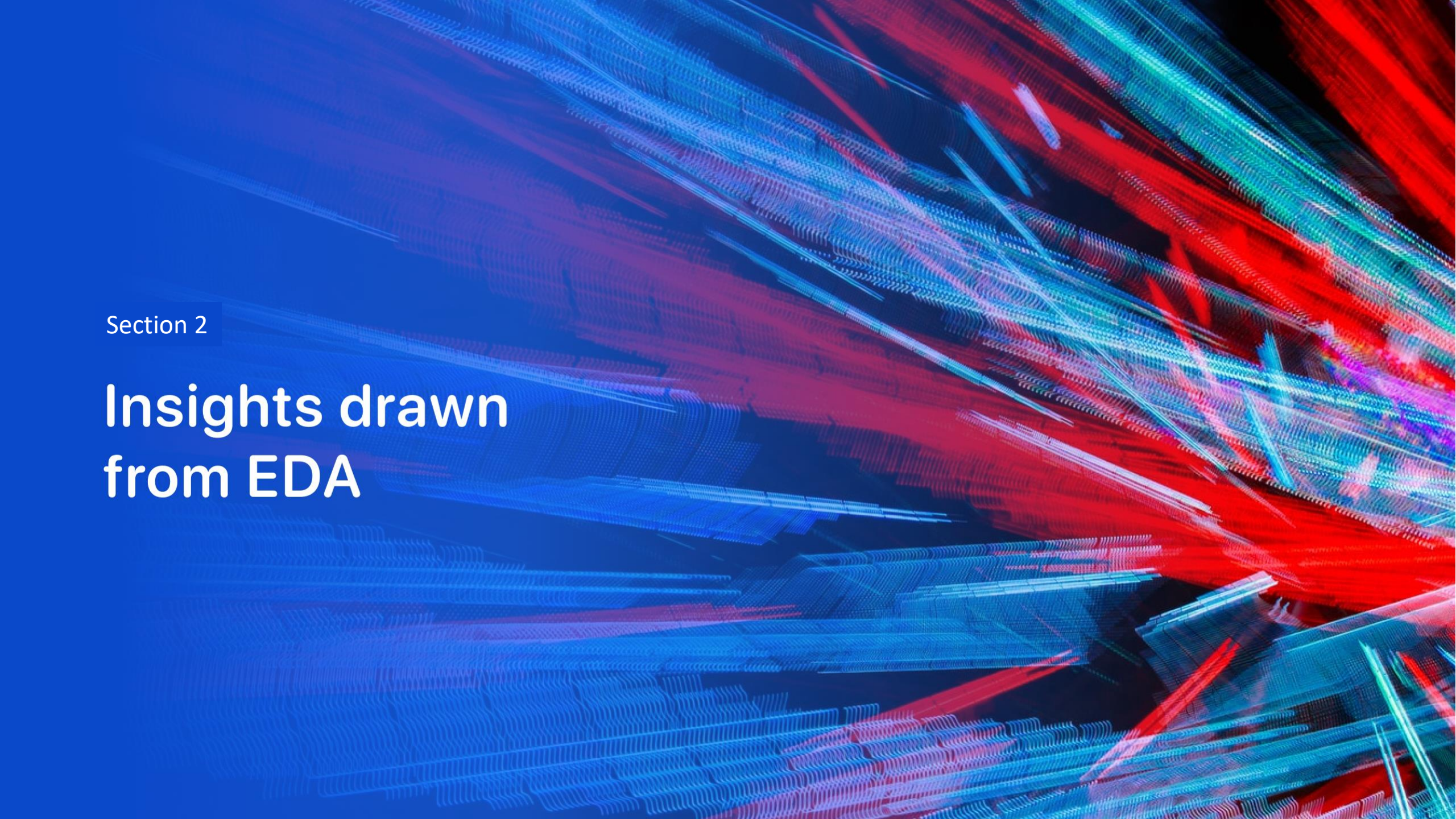
- I used this flowchart of the model development process :
- GitHub URL of the notebook is below :

https://github.com/EladdadMahdi/IBM-Applied-Data-Science-Capstone/blob/dev/notebooks/Module_04/7_Machine_Learning_Prediction.ipynb



Results

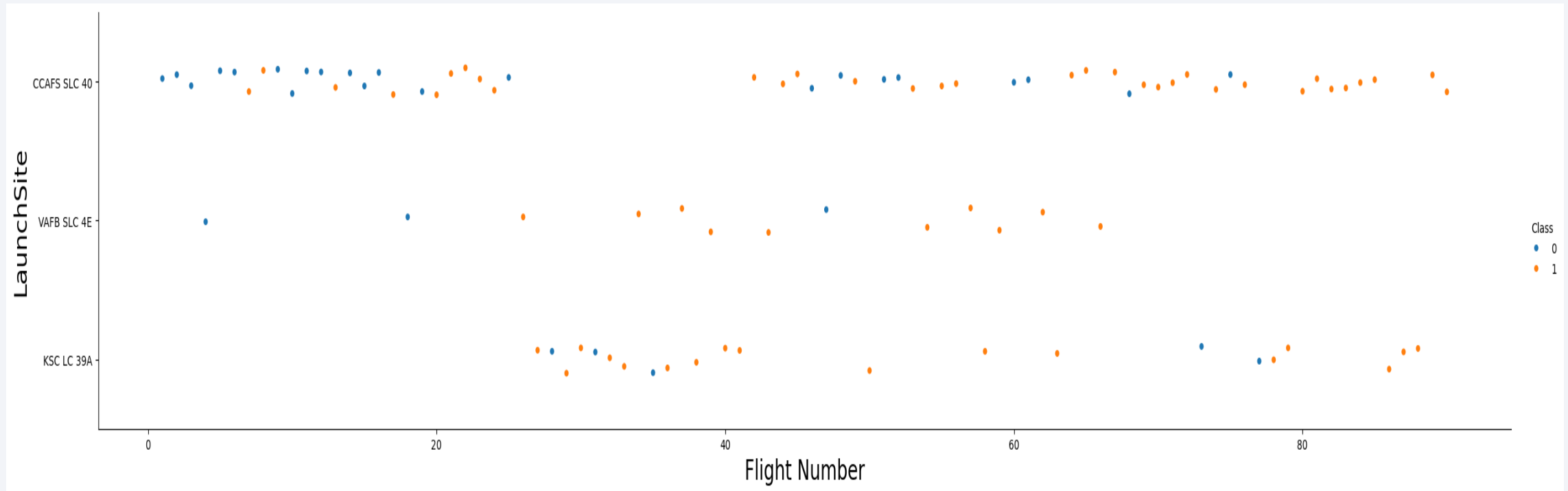
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is dynamic and technological.

Section 2

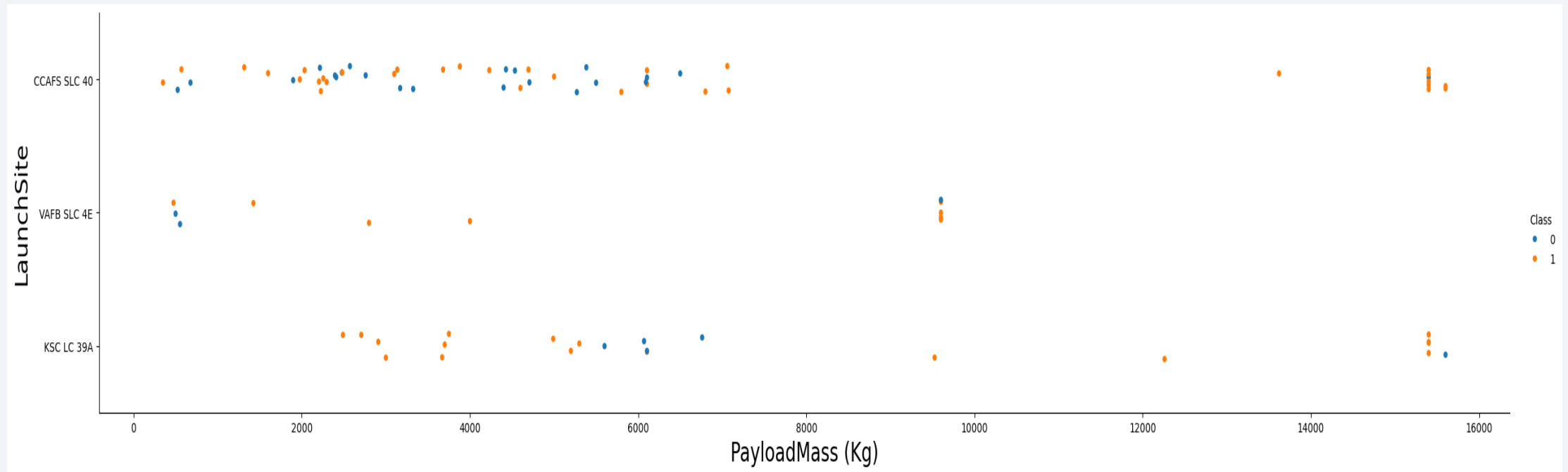
Insights drawn from EDA

Flight Number vs. Launch Site



- SpaceX's first-stage landing success rate improved dramatically with flight experience (Flight Number) → iterative engineering learning.

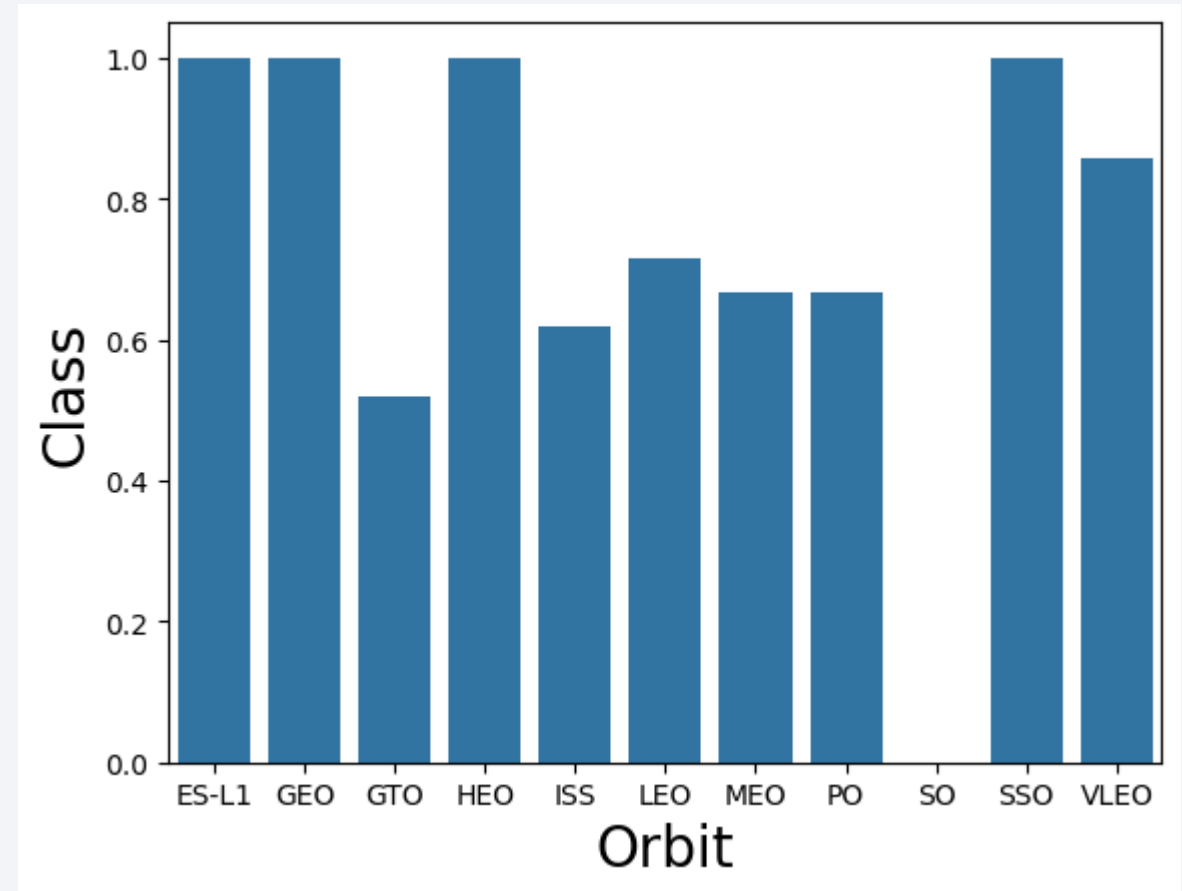
Payload vs. Launch Site



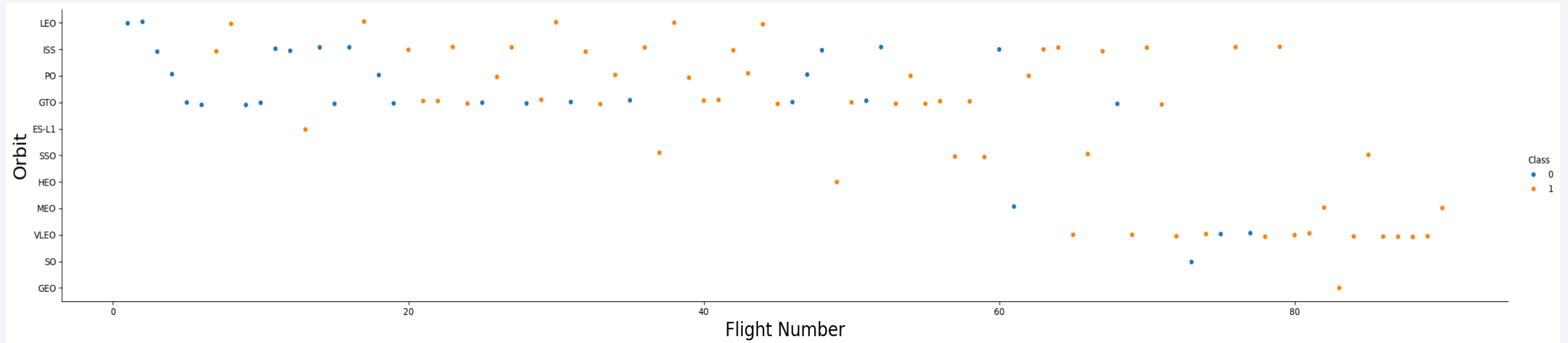
- Launch site determines payload capability and success rate.

Success Rate vs. Orbit Type

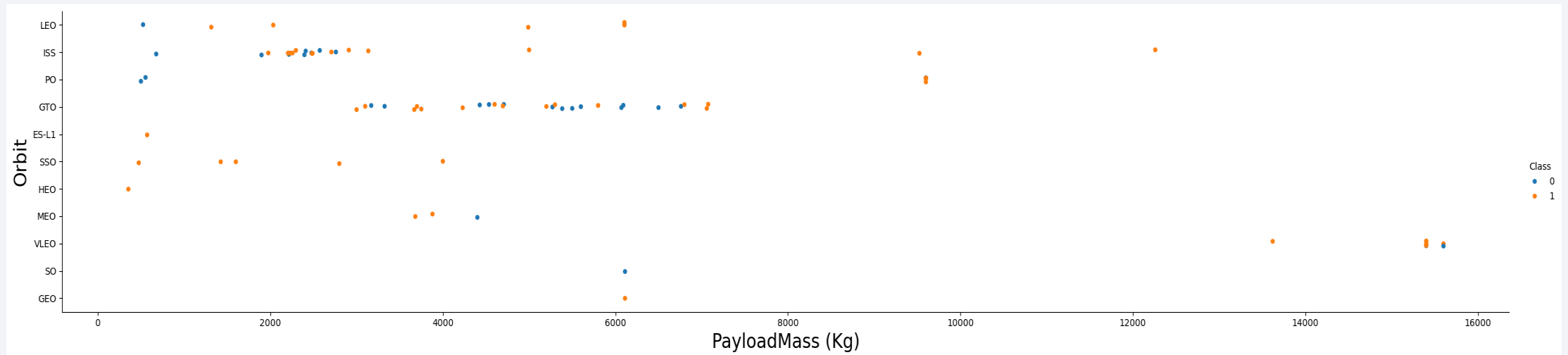
- ES-L1, GEO, HEO, SSO, and VLEO orbits show the highest first-stage landing success rates.
- GTO, PO, MEO orbits show the lowest first-stage landing success rates.



Flight Number vs. Orbit Type



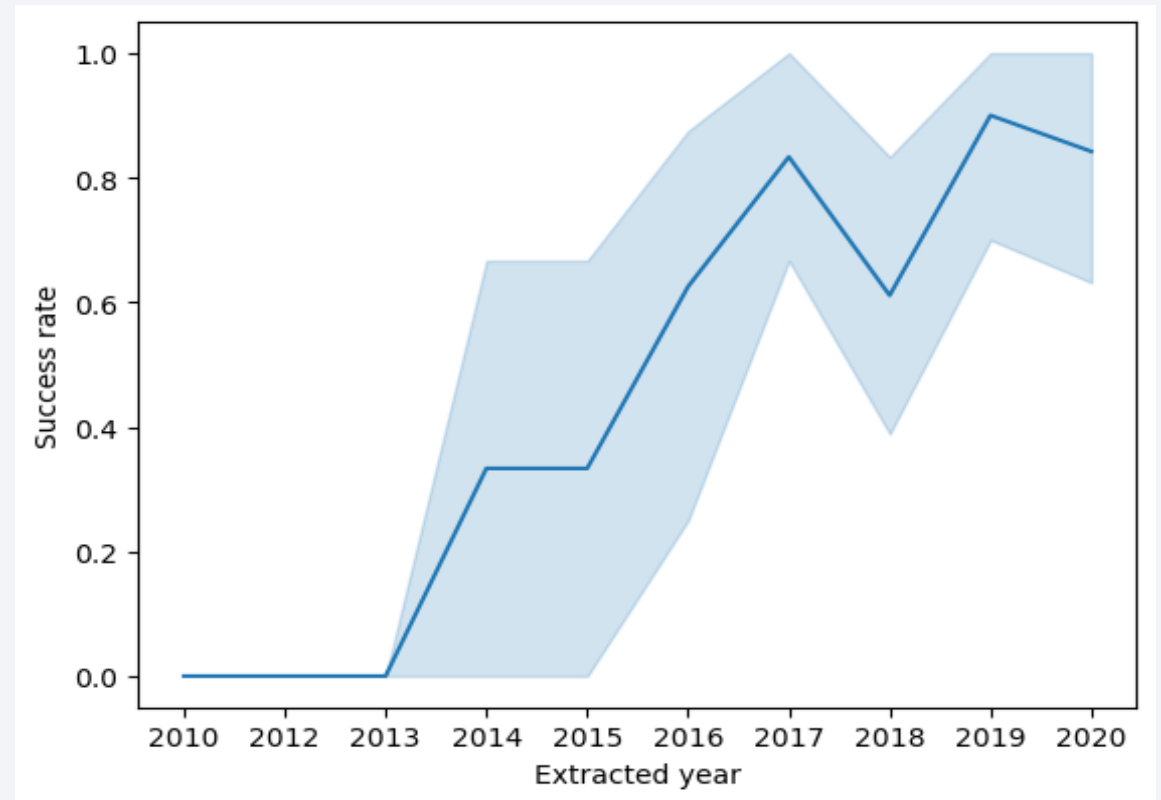
Payload vs. Orbit Type



- heavy payloads (>10k kg) successful landings remain dominant in LEO/PO/ISS.

Launch Success Yearly Trend

- This plot shows Falcon 9 first-stage landing success rates rising sharply from 2013 until 2020.



All Launch Site Names

- I find the names of the unique launch sites using Jupyter's %sql magic command. The query applies the **DISTINCT** command on the 'LaunchSite' column to select unique names of launch sites from the SPACEXTBL table.

```
Task 1

Display the names of the unique launch sites in the space mission

%sql select distinct Launch_site from SPACEXTBL;

* sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

Task 2

Display 5 records where launch sites begin with the string 'CCA'

```
%sql select * from SPACEXTBL where Launch_site like 'CCA%' limit 5
```

```
* sqlite:///my_data1.db  
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- I display 5 records where launch sites begin with 'CCA' using Jupyter's %sql magic command. The query uses LIKE 'CCA%' pattern matching on the 'LaunchSite' column from SPACEXTBL, limited to 5 rows with **SELECT**.

Total Payload Mass

```
Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

%sql select sum(PAYLOAD_MASS__KG_) as Total_payload from SPACEXTBL where Customer like '%NASA (CRS)%';

* sqlite:///my_data1.db
Done.

Total_payload
48213
```

- I calculated the total payload mass carried by boosters launched for NASA CRS using Jupyter's %sql magic command.
- The query applies SUM(PAYLOADMASS__KG) aggregate function on the payload column from SPACEXTBL table, filtered by Customer LIKE '%NASA CRS%' to match NASA missions.
- Result: 48,213 kg total from ~20 early CRS missions to ISS (2012-2018).

Average Payload Mass by F9 v1.1

Task 4

Display average payload mass carried by booster version F9 v1.1

```
%sql select avg(PAYLOAD_MASS_KG_) as Average_payload_mass from SPACETABLE where Booster_Version = 'F9 v1.1';
```

```
* sqlite:///my_data1.db  
Done.
```

Average_payload_mass
2928.4

- I display the average payload mass carried by booster version F9 v1.1 using Jupyter's %sql magic command.
- The query applies AVG(PAYLOADMASS__KG) aggregate function on the payload column from SPACETABLE, filtered by BoosterVersion = 'F9 v1.1' to select missions using this early Falcon 9 variant.“
- Result: 2,928.4 kg average from dataset missions.

First Successful Ground Landing Date

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint: Use min function

```
%sql select date as First_Successfull_landing_date from SPACEXTBL where Landing_Outcome like '%Success (ground pad)%' order by date asc limit 1;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
First_Successfull_landing_date
```

```
2015-12-22
```

- I list the date of first successful landing on ground pad using Jupyter's %sql magic command. The query selects Date AS First_Successful_landing_date from SPACEXTBL, filtered by Landing_Outcome **LIKE** '%Success ground pad%', ordered by Date ASC with **LIMIT 1** to get the earliest success."
- Result: 2015-12-22—first Falcon 9 ground pad recovery milestone.

Successful Drone Ship Landing with Payload between 4000 and 6000

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
%sql select Booster_Version as Booster_Name from SPACEXTBL where Landing_Outcome like '%Success (drone ship)%' and PAYLOAD_MASS_KG_ between '4000' and '6000';
```

```
* sqlite:///my\_data1.db
```

```
Done.
```

Booster_Name

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

- I filtered drone ship successes with 4000 kg-6000 kg payloads and return booster names. I used the **WHERE** clause with key commands **LIKE** for pattern match and **AND** for multiple payload conditions.

Total Number of Successful and Failure Mission Outcomes

Task 7

List the total number of successful and failure mission outcomes

```
%sql select count(Mission_Outcome) as Total_number_of_mission_outcomes_success from SPACEXTBL where Mission_Outcome like '%success%';
```

```
* sqlite:///my_data1.db  
Done.
```

Total_number_of_mission_outcomes_success
100

```
%sql select count(Mission_Outcome) as Total_number_of_mission_outcomes_failure from SPACEXTBL where Mission_Outcome like '%failure%';
```

```
* sqlite:///my_data1.db  
Done.
```

Total_number_of_mission_outcomes_failure
1

- I calculate the total number of successful and failed mission outcomes using Jupyter's %sql magic command. The first query counts MissionOutcome filtered by **LIKE '%success%'** to get **100 successes**.
- The second query uses the same approach with **LIKE '%failure%'** to get **1 failure**.

Boosters Carried Maximum Payload

- I list all booster versions that carried the maximum payload mass using a subquery. First finds 15,600 kg max payload. Second selects matching F9 B5 Block 5 boosters.
- Results: 10+ F9 B5 boosters carried 15,600 kg max.
- Key commands: MAX(), subquery.

Task 8

List all the booster_versions that have carried the maximum payload mass. Use a subquery.

```
%sql select max(PAYLOAD_MASS_KG_) as Maximum_payload_mass from SPACEXTBL;
```

```
* sqlite:///my_data1.db  
Done.
```

Maximum_payload_mass
15600

```
%sql select Booster_Version, PAYLOAD_MASS_KG_ from SPACEXTBL where PAYLOAD_MASS_KG_ = (select max(PAYLOAD_MASS_KG_) from SPACEXTBL) ;
```

```
* sqlite:///my_data1.db  
Done.
```

Booster_Version	PAYLOAD_MASS_KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

2015 Launch Records

Task 9

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

```
%sql select substr(Date,6,2) as month, Landing_Outcome,Booster_Version, Launch_site from SPACEXTBL where substr(Date,0,5)='2015' and Landing_Outcome ='Failure (drone ship)';
```

```
* sqlite:///my\_data1.db
```

```
Done.
```

month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

- find all 2015 missions that failed to land on drone ships. I display the month, landing outcome, booster, and launch site for each failure."
- Key commands: SUBSTR() (extract month/year from date), WHERE, AND (filter 2015 + failure).

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- I rank all landing outcomes by how many times they occurred between 2010-06-04 and 2017-03-20, sorted from most to least frequent. Query counts each outcome type and orders results in descending order.
- Key commands: COUNT() (count occurrences), GROUP BY (group by outcome type), ORDER BY DESC (highest to lowest), BETWEEN (date range filter).

Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

```
%sql select Landing_Outcome, count(Landing_Outcome) as Landing_outcome_count from SPACEXTBL where Date between '2010-06-04' and '2017-03-20' group by Landing_Outcome order by Landing_outcome_count desc;
```

* [sqlite:///my_data1.db](#)

Done.

Landing_Outcome	Landing_outcome_count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

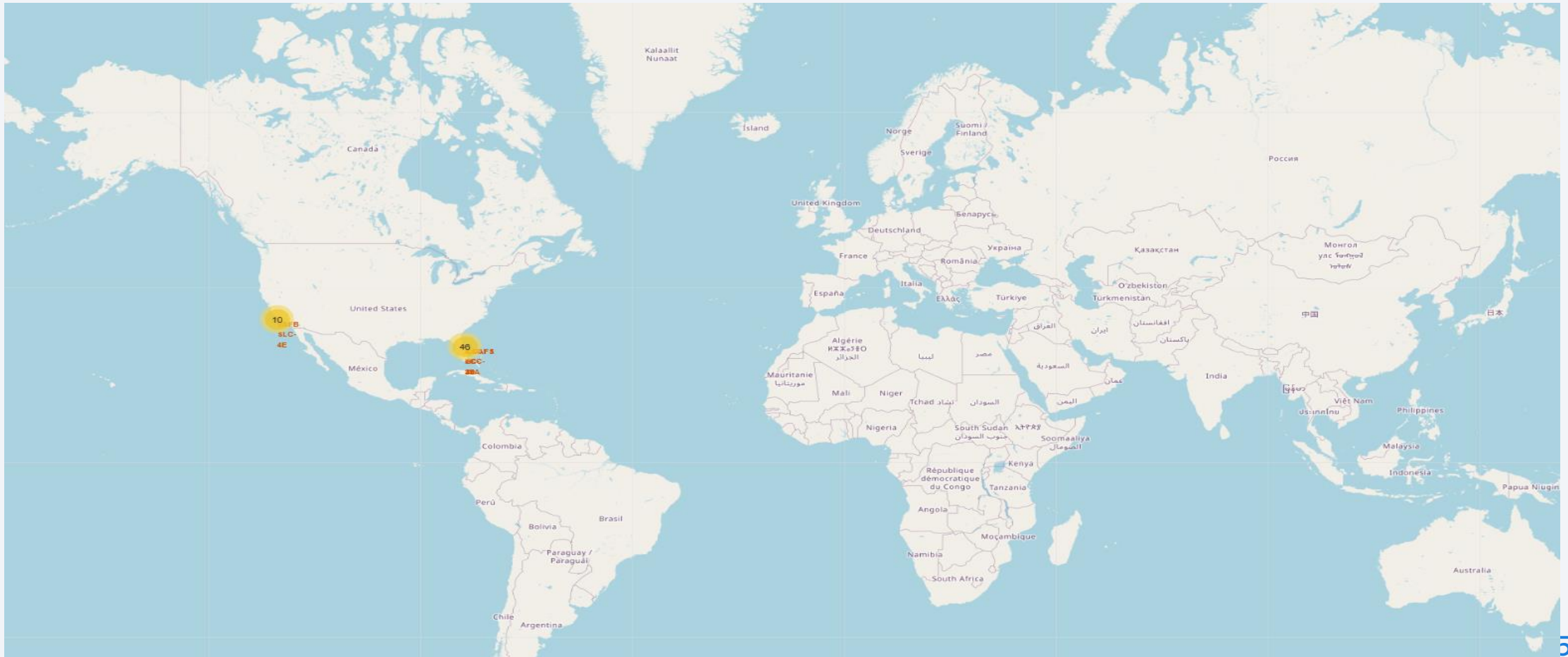
A satellite view of Earth from space, showing the curvature of the planet and the glowing lights of cities and continents against the dark background of space. The Earth's surface is a mix of dark blue oceans and lighter blue/white landmasses, with numerous bright yellow and orange lights indicating urban areas.

Section 3

Launch Sites Proximities Analysis

Global Distribution of SpaceX Launch Sites

SpaceX launch sites are strategically located along USA coasts, including the Atlantic (Florida) and Pacific (California) shorelines.

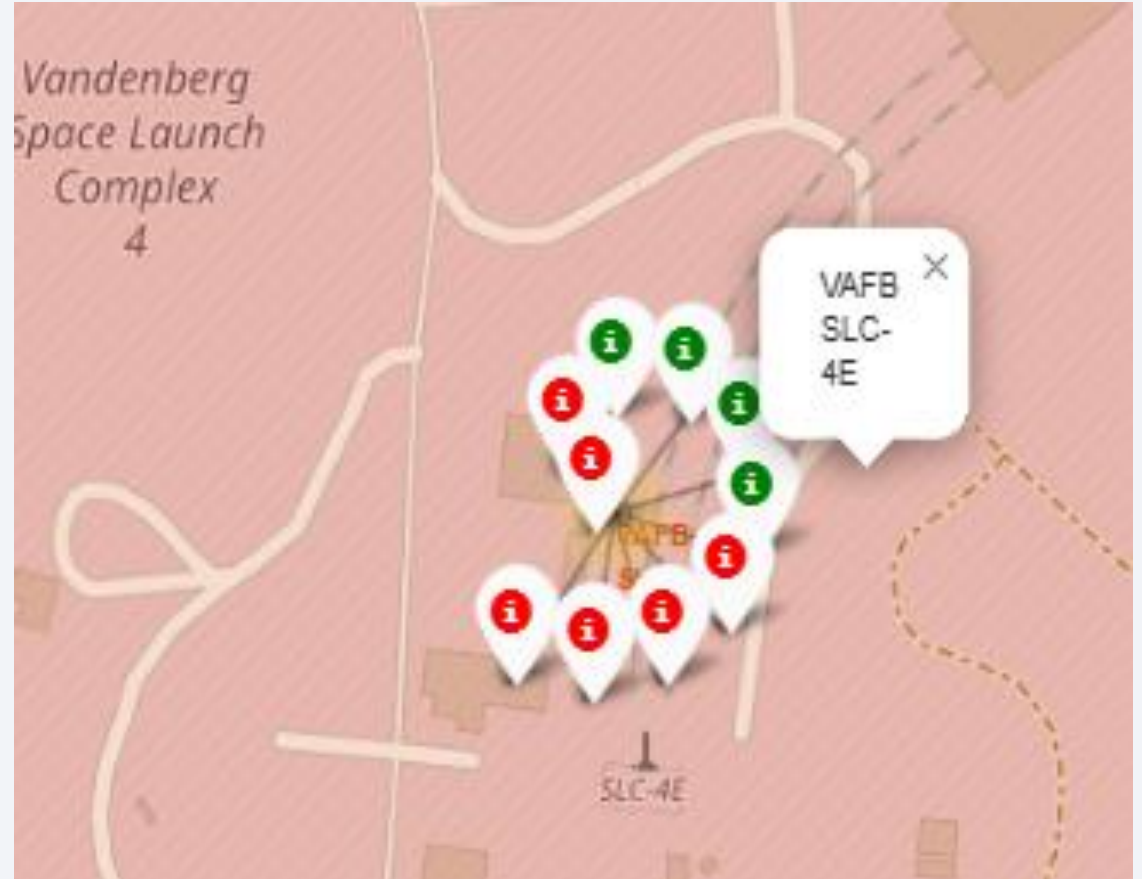


Labeled Launch Sites Markers

NB : Green Markers are for successful launches and Red Markers are for failed launches

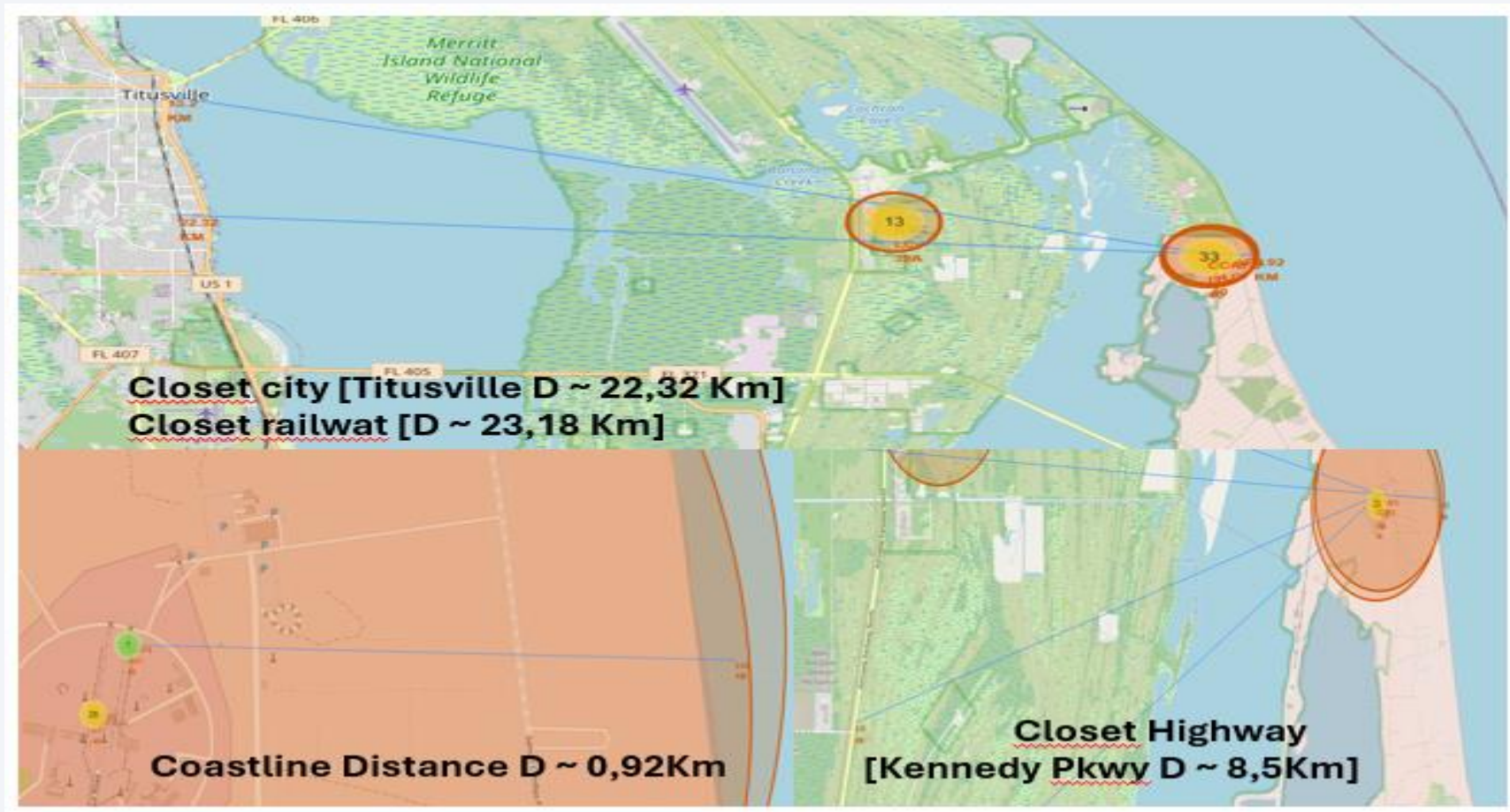


Florida launch sites



California launch sites

Distances between landmarks and launch sites



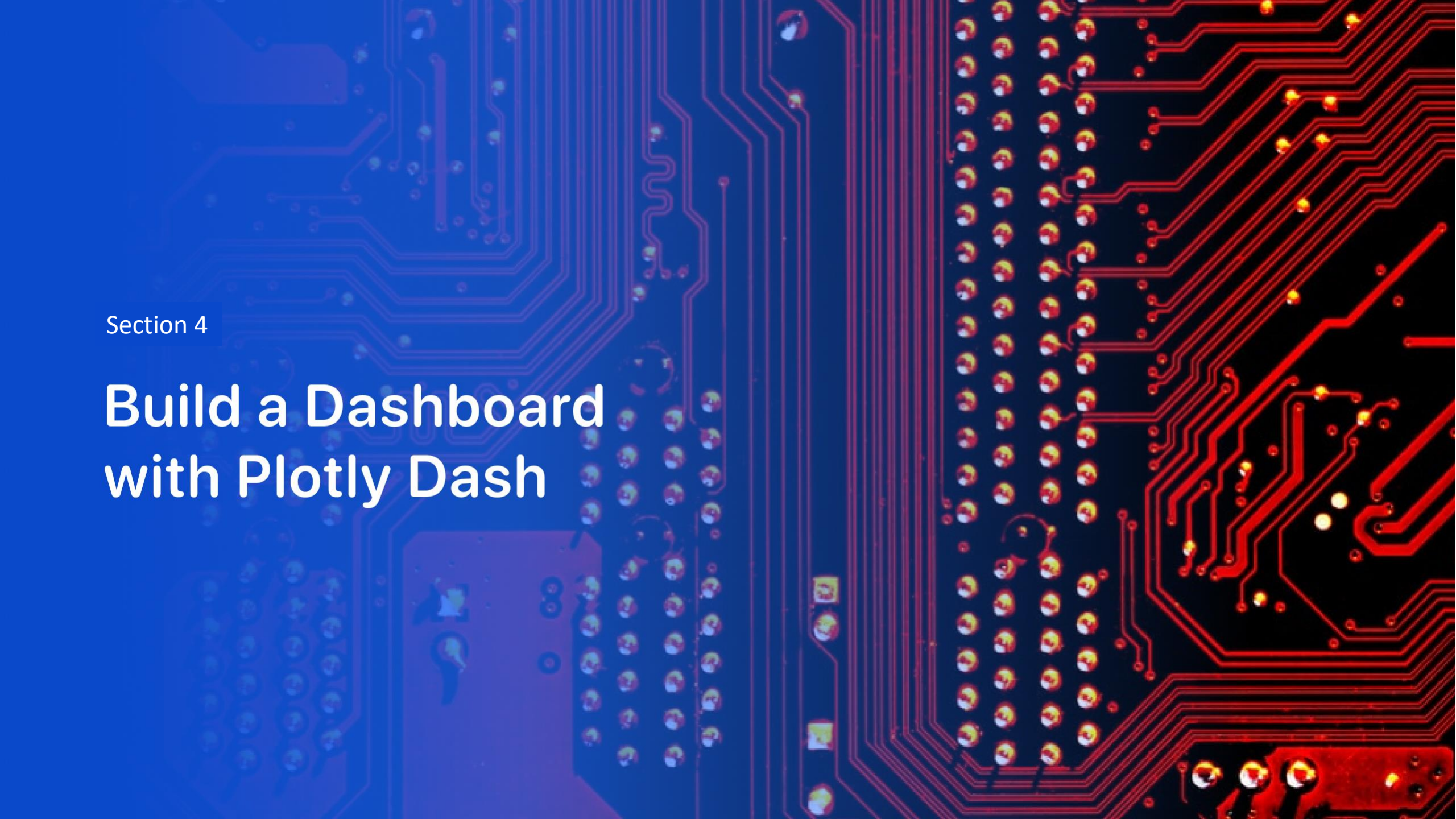
Distances between landmarks and launch sites

NB : Green Markers are for successful launches and Red Markers are for failed launches

- *Are launch sites in close proximity to railways?*
→ *No. Railway at 23 km*
- *Are launch sites in close proximity to highways?*
→ *No, Highway at 8.5 km*

Are launch sites in close proximity to coastline?
→ *Yes. Coastline at 0.92 km -- intentionally coastal for safety*

- *Do launch sites keep certain distance away from cities?*
- → *Yes. City at 22 km -- protection of civilians.*

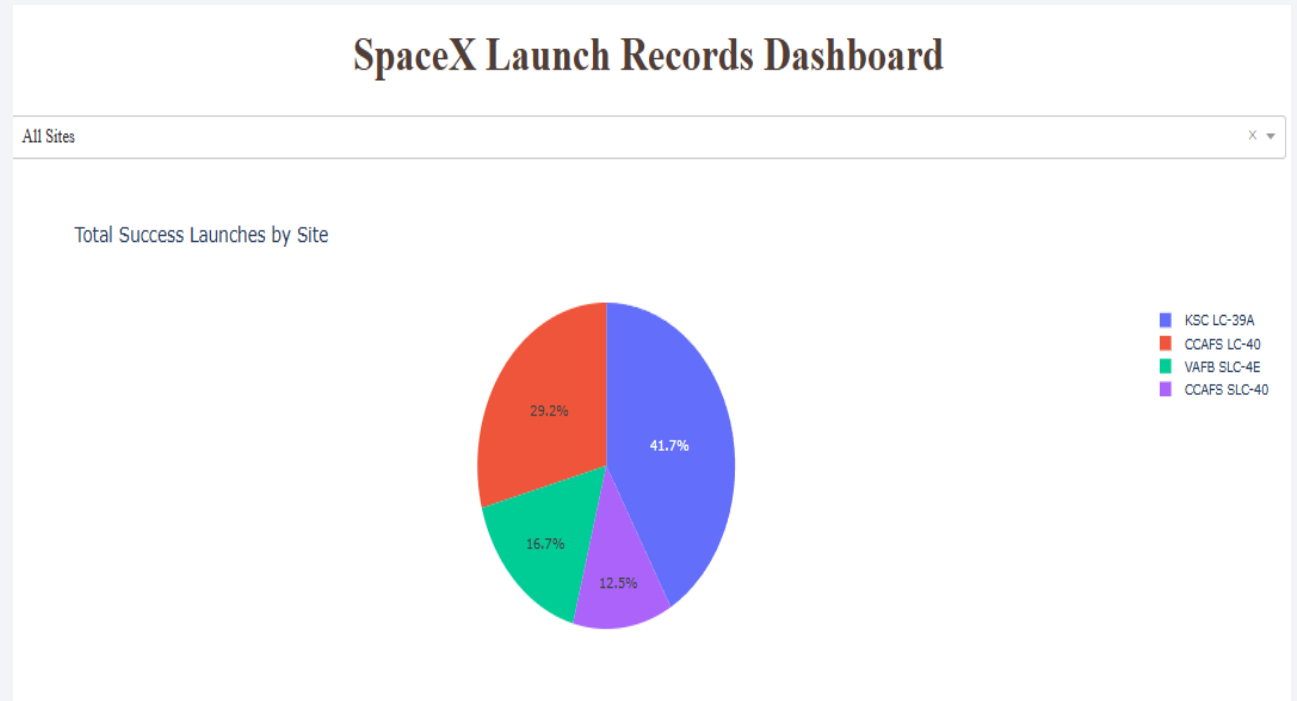


Section 4

Build a Dashboard with Plotly Dash

Success percentage of each launch site

- *KSC LC-39A dominates with 41.7% of successful launches*
- *Cape Canaveral pads combined also hit 41.7%.*



Launch site [KSC LC-39A] with the highest launch success rate

SpaceX Launch Records Dashboard

KSC LC-39A

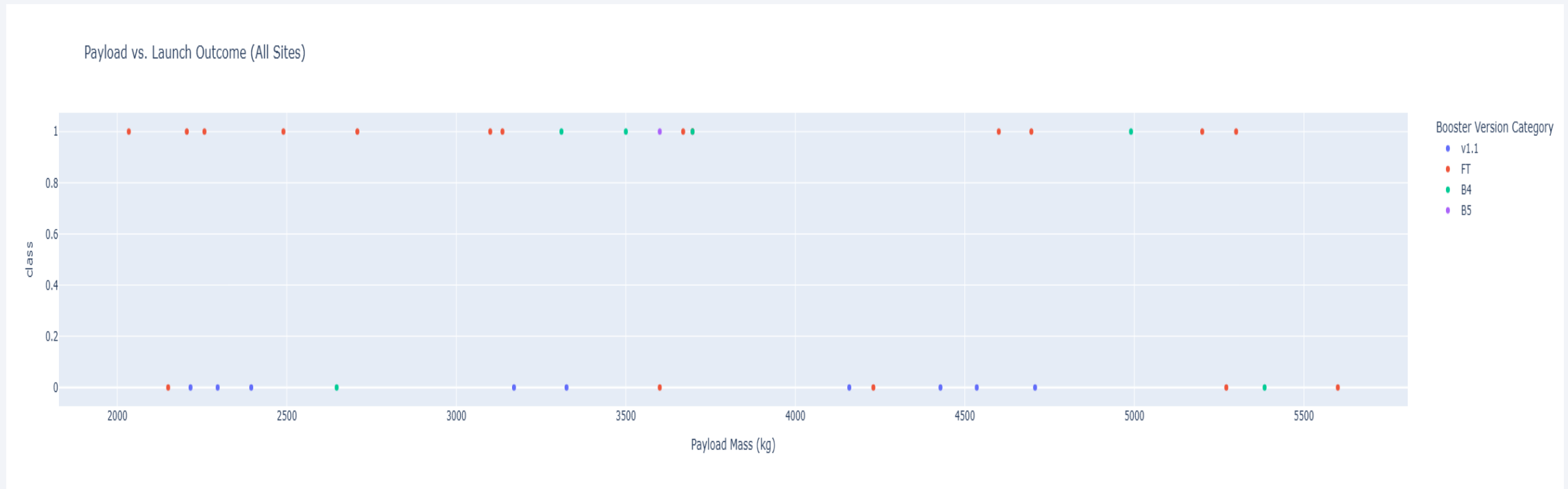
Launch Success Rate for KSC LC-39A

- *KSC LC-39A → 76,9% while 23,1% as failure rate.*



Payload vs Launch Outcome for all sites, with different payloads [5000Kg/10000Kg]

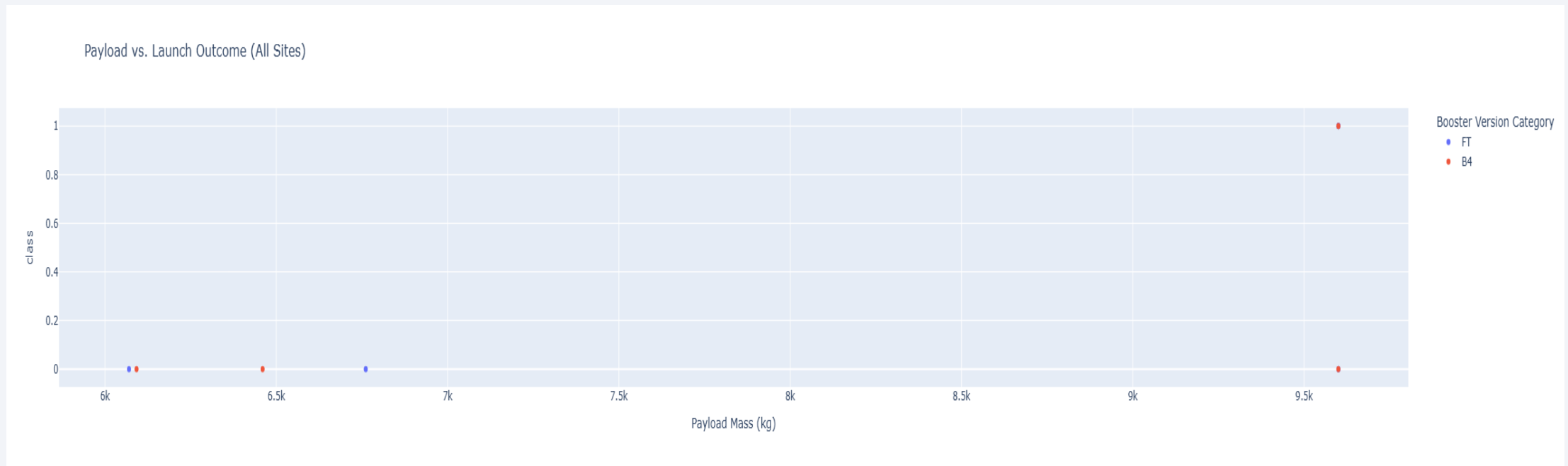
[2000-6000Kg]payload range



- *[2000-6000]Kg payload range → highest launch success rate*
- *FT/B4/B5 boosters achieves successful launches at this payload range*

Payload vs Launch Outcome for all sites, with different payloads [5000Kg/10000Kg]

[6000-10000Kg]payload range



- *[6000-10000]Kg payload range → lowest launch success rate*
- *Only B4 booster achieves successful launches at this payload range*

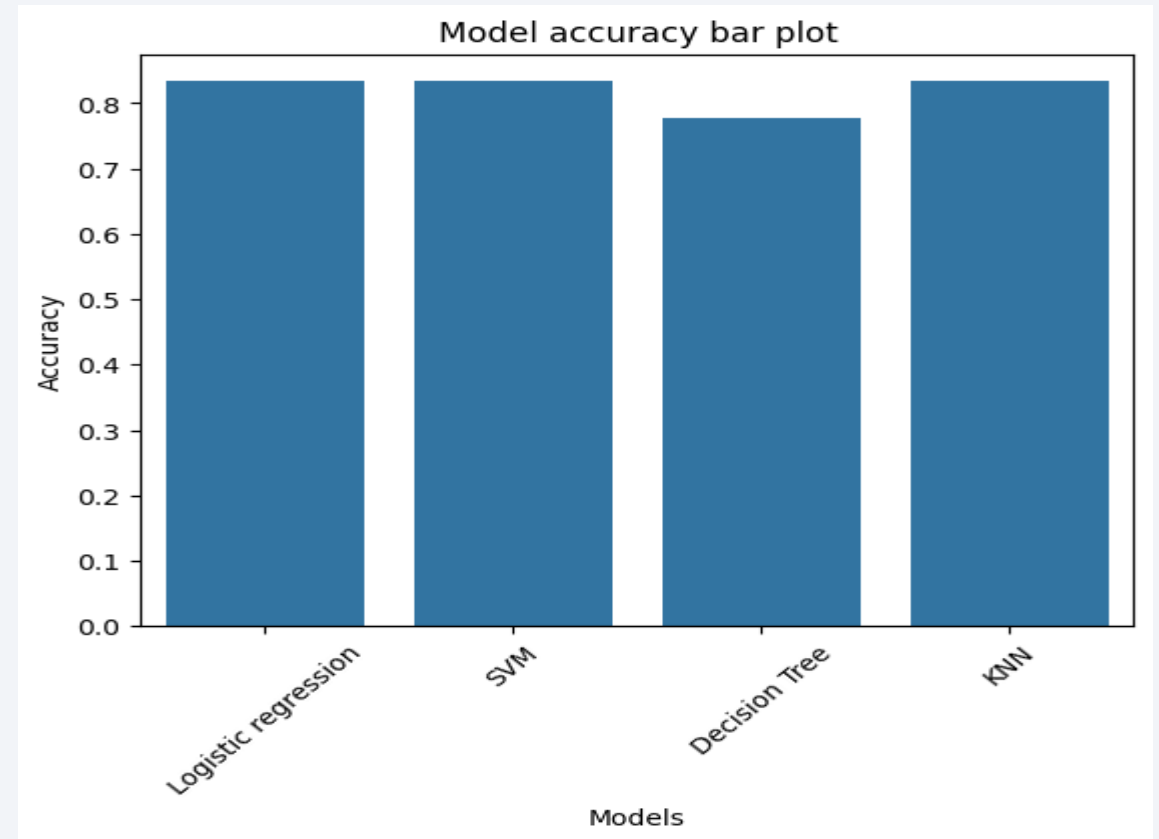


Section 5

Predictive Analysis (Classification)

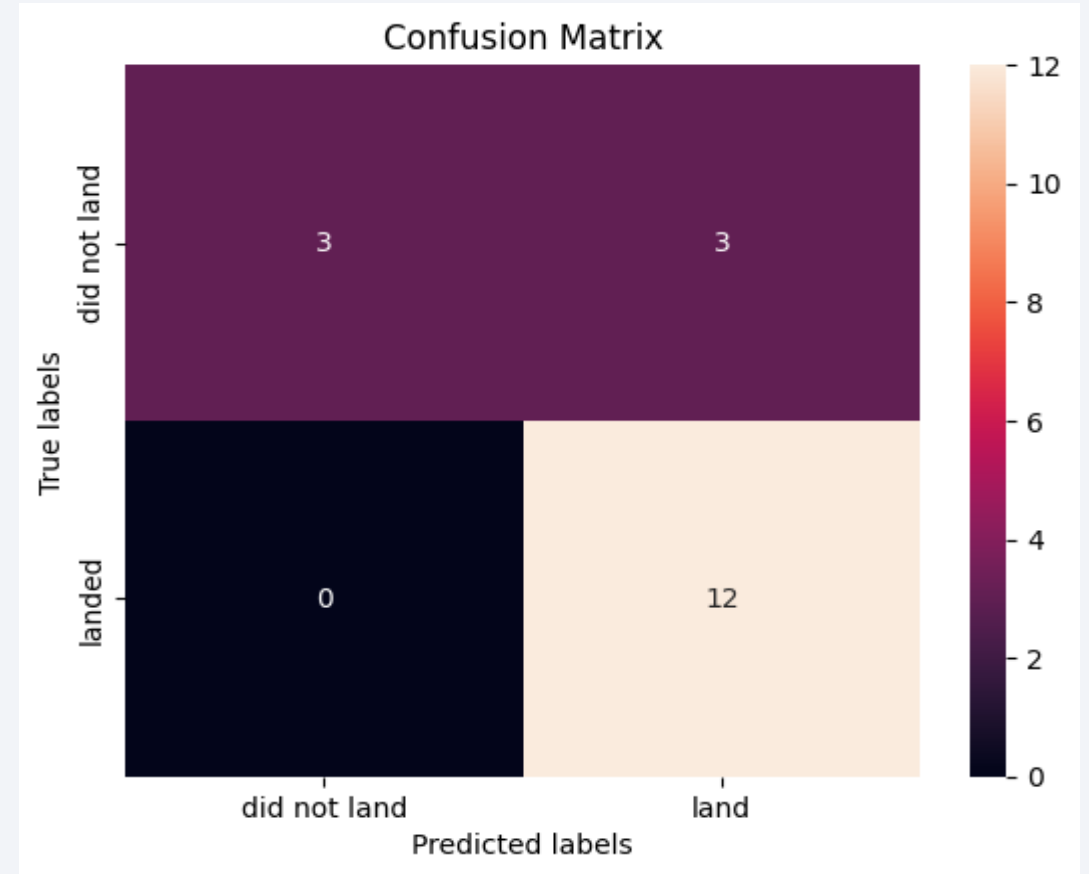
Classification Accuracy

- Bar chart to Visualize the built model accuracy for all built classification models
- Best performing model is :
SVM (accuracy of 83%)



Confusion Matrix

- confusion matrix of the best performing shows that the model is struggling with False Positives (FP) it predicted 3 successful landing while they are not.



Conclusions

- **Key Factors Influencing Landing Success:**
 - Launch site (KSC LC-39A: 76.9%), payload range (2000-6000 kg optimal)
 - Orbit type (ES-L1/GEO/SSO highest; GTO lowest)
- **Feature Interactions Affecting Success:**
 - Heavy payloads (>10,000 kg) succeed primarily in LEO/ISS orbits
 - GTO missions show low success regardless of flight number
 - Later boosters (B4) handle wider payload ranges successfully
- **Success Improves with Experience:**
 - Dramatic improvement 2013-2020; later flight numbers correlate with success
 - Coastal sites (0.92 km from shore) with 22 km buffer from cities enable recovery
- **Model Performance:**
 - SVM algorithm: 83% accuracy with 3 false positives

Thank you!

